Ishaan Salian

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Experience

Praan Inc. **Mumbai, India**
Embedded Systems Engineer **August 2026 - Present**

- Designed and integrated the NFC-based automated filter detection system, enabling under-2-minute filter servicing and transparent filter-usage reporting in the companion app, addressing a customer trust gap around filter life
- Consolidated separate V1/V2/V3 firmware branches into a single release image with runtime hardware-revision detection and feature gating, so a single OTA release delivers connectivity, reliability, and security updates across the installed base
- Built automatic WiFi/Bluetooth failover into the device's connectivity architecture, eliminating connectivity blackouts for end users; implemented secure BLE and SoftAP-based provisioning as alternatives to WiFi-only setup, and root-caused recurring BLE/WiFi/MQTT lifecycle defects, supporting a successful global product launch
- Redesigned the OTA update flow with downgrade protection, post-update self-confirmation, and automatic rollback to the last verified firmware on the next power cycle, reducing field-update failure risk
- Leading the second-generation PCB revision for the product's ESP32-based control board, translating updated sensor and manufacturing requirements from mechanical and production teams into a production-ready design
- Developed a fan calibration test utility enabling live, incremental duty-cycle tuning without per-configuration firmware builds, streamlining production validation across fan batches

Dynamic and Autonomous Robotic Systems Laboratory **Amherst, MA USA**
Robotics Research Engineer **November 2025 - July 2026**

- Designed, assembled, and brought up 6-layer mixed-signal PCB stackups for the Dash humanoid robot, including dedicated power/ground planes, low-inductance return paths, and CAN routing for high-current robotic actuators
- Redesigned the 48V (400A peak) power distribution system for the PresToe humanoid robot, incorporating improved protection circuitry, transient suppression, Hall-effect current sensing, and CAN telemetry to increase system safety

Projects

Autonomous Workspace Organizer Robot | *KiCad, Fusion 360, BLE, Object Detection* **Senior Design Project**

- Built vision-guided desk-organizing robot by leading the mechatronic development of a 4-layer ESP32-S3 control PCB with onboard 5V converter and BMS, and modeling compact tank-track chassis driven by continuous rotation servos
- Implemented closed-loop position control where an overhead Jetson Nano runs YOLOv8 object detection and sends movement commands over BLE, using the robot's ArUco marker to correct for positioning drift and servo backlash

FreeRTOS based Multi-Sensor Data Logger | *ESP32, FreeRTOS*

- Developed FreeRTOS-based system with 5 concurrent tasks managing multi-rate sensor acquisition (100Hz, 1Hz, 0.5Hz) with mutex-protected shared I2C bus; implemented real-time sensor data logging to an SD card for behavior analysis
- Debugged priority inversion issue where the low-priority environmental sensor task held the I2C mutex during blocking reads, causing the high-priority IMU task to miss its sampling deadline and drop data

Mini Golf Robot | *Ranked 5th out of 20+ Teams* **UMass Mechatronics Design Team (ASME)**

- Co-led electronics subteam for competition robot, integrating dual NEMA23 stepper motors with M542C precision drivers to achieve torque-controlled putting and chipping swings
- Designed wireless control system over bluetooth, mapping joystick inputs to drivetrain and buttons to swing actuation

Education

University of Massachusetts Amherst **Amherst, MA USA**
Bachelor of Science in Computer Engineering **Graduated May 2025**

Leadership & Teaching

Undergraduate Teaching Assistant | 4 ECE Courses **September 2024 - May 2025**

- Assisted in Physical Computing, ECE Junior Design, and Security Engineering courses; supporting 250+ students through bare-metal C programming, hardware debugging, and secure embedded system design

Technical Skills

Embedded Development: C, ESP32/STM32/nRF52, FreeRTOS, interrupt-driven programming
Hardware Design: KiCad, oscilloscope debugging, I2C/SPI/CAN/UART, BLE, soldering/rework
Software: Python, C++, MATLAB, Bash scripting, Git, Linux, Fusion 360, OpenCV